



Machine Learning Model for Automated Feedback on Video Intubations

BENG 493 - Senior Advanced Design Project II

Final Report

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Executive Summary

Endotracheal intubation is a critical medical procedure used to secure a patient's airway when they cannot breathe effectively on their own or require airway protection. Fairfax County Fire and Rescue currently reviews video-assisted intubation recordings manually to provide feedback to trainees. While manual review supports training quality, it requires significant time from experienced clinicians and can delay feedback. The goal of this project is to develop a machine learning system that supports the review process by providing an automated first-pass classification of video-assisted intubation attempts as PASS or FAIL.

This project used recorded GlideScope videos from simulated intubation attempts performed on an adult manikin at ASTEC. The videos were labeled with key anatomical structures, including the vocal cords, arytenoids, esophagus, and endotracheal tube. These labels were used to train and evaluate machine learning models that identify visual patterns associated with successful and unsuccessful intubation attempts. The system is not intended to replace clinician judgment. Instead, it is designed to help educators prioritize videos that need closer review and reduce the manual screening burden.

The final design uses a supervised 2D CNN + SVM hybrid model. The 2D convolutional neural network extracts spatial features from video frames, while the support vector machine uses those extracted features to classify each full video as PASS or FAIL. This design was selected because it provided the strongest balance of classification performance, training efficiency, and suitability for the available dataset. The team also began prototyping a Mask R-CNN anatomy detector to support future feedback beyond binary classification.

Three tests were performed to verify and validate models. The first test compared two shapes (rectangles and polygons) that were used to annotate video data. Rectangle shapes were found to be more efficient and suitable for binary classification tasks, while polygon shapes better defined the locations of anatomical structures for subsequent detection. The second test compared the performance of 3D CNN and 2D CNN + SVM across different image sizes. The 2D CNN + SVM achieved acceptable levels of performance as measured by accuracy (83.38% to 83.81%), F1 score of 0.843, and AUC (area under the curve) of 0.926 at all tested resolutions. The second test also showed a shorter average training time of 18-23 seconds when compared with the 3D CNN. Test three evaluated model performance of both model designs across 5 resolutions between 64x64 and 600x600. The final test compared model output against the clinician's evaluation of model output. Results from this test show that clinicians generally agree with the model output. However, there were a few videos that led to disagreement among clinicians, indicating that there is still a need for a standardized way to review and interpret video data.

Overall, the final prototype met the main technical requirements of accuracy above 80%, F1-score above 0.80, efficient processing, secure data handling, and interpretable PASS/FAIL output. The main limitations are the small dataset, single-manikin testing environment, limited clinician responses, and binary-only output. Future work should expand the dataset, collect more clinician evaluations, and integrate the completed anatomy detector to provide more detailed training feedback.

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Problem Statement

Endotracheal intubation (ETI or EI) is a medical procedure commonly performed in emergency settings by medical providers, including paramedics working within local fire departments. The intubation procedure is performed on patients who are incapable of breathing on their own, whose airway requires clearance of obstructions, or protection from the inhalation of fluids [1]. A successful intubation secures the airway by isolating the oral cavity from the bronchi and creates a direct path to the patient's lungs for ventilation. The procedure is not without risks; prolonged or difficult intubations can worsen the patient's condition by causing trauma to the larynx and oral cavity. Broken teeth, lacerations to the vocal cords or the trachea, bleeding, infection, and swelling are potential risks associated with EI [1].

Video intubation, also known as video-assisted EI or video laryngoscope, differs from traditional EI due to the implementation of a camera attached to the laryngoscope—the rigid tool used to expose the airway—which transmits the image to a monitor for enhanced visualization and airway navigation during the procedure. Fairfax County Fire and Rescue (FCFRD), a local fire department, trains emergency responders using recordings of video intubation procedures and providing feedback accordingly, which can take a great deal of time and effort to complete. We aim to reduce variability and manual labor required for EI training by implementing a machine learning (ML) model, a subset of artificial intelligence (AI), to generate automated feedback and categorization, effectively serving as an initial review for educators, with the long-term goal of improving patient outcomes.

Background/Relevance

An alternative procedure to EI is laryngeal mask airway, which is less invasive and is associated with less discomfort and risks for the patient. However, EI offers superior airway protection, extended duration of use, and more precise ventilation control. EI is more often used in high-risk, emergency cases where there is an increased risk of complications, further highlighting the importance of efficiency and accuracy in intubations [2]. A few common errors that can occur during EI are poor positioning of the laryngoscope blade impeding visualization and tube navigation, incorrect endotracheal tube placement, and excessive force. Proper training on the procedure is essential for reducing the occurrence of such errors.

Fig.1 shows a diagram of a direct, or traditional, intubation where the person performing the procedure is visualizing the vocal cords and airway by directly examining the patient's larynx (throat), as opposed to video intubation where a camera and monitor are used. Fig. 1A shows the laryngoscope blade in the oral cavity pressing against the patient's tongue, and Fig. 1B shows the endotracheal tube being inserted between the patient's vocal cords into their airway by applying gentle force to the stylet inserted in the tube. Clear visualization of the airway is essential for endotracheal (ET) tube navigation to prevent collisions between the tube and sensitive larynx structures (e.g., epiglottis), insertion of the tube into the esophagus—a passage to the stomach—and endobronchial intubation, where one lung gets ventilated while the other does not.

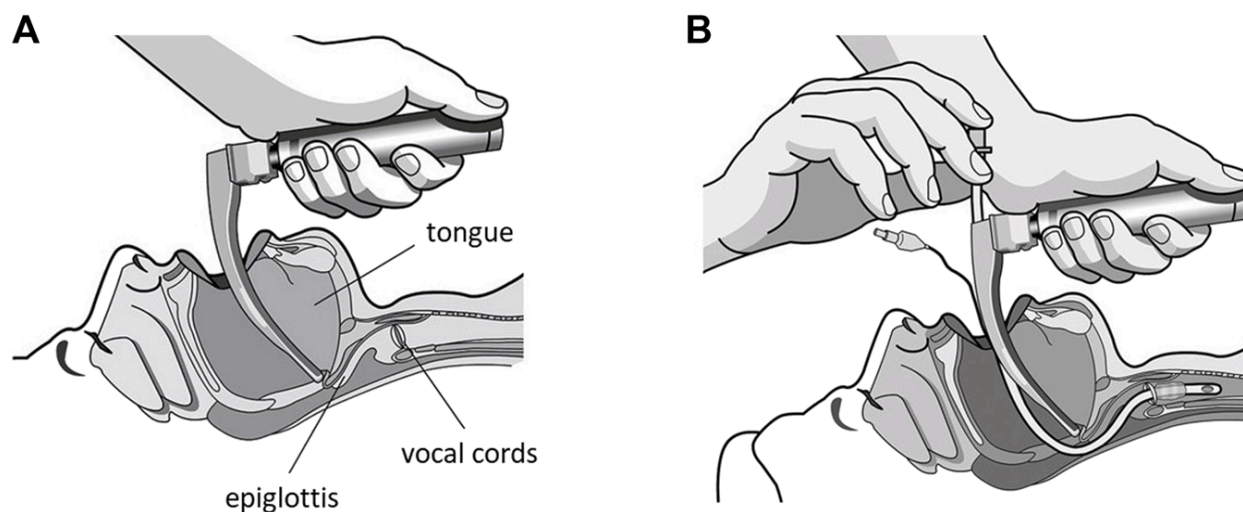


Figure 1. Endotracheal intubation procedure.

The FCFRD manually reviews video intubations performed by trainees to provide feedback and coaching. Assistant Operational Medical Director at FCFRD, Dr. Dan Avstreich, has stated the time and energy commitment required to review these videos causes the training process to be deferred or postponed for more pressing matters, such as suppressing fires or participating in search and rescues. Machine learning offers a unique opportunity of reducing time and effort required by trainers and educators. ML is a subset of AI where patterns are

identified and stored (“learned”) from training data to make conclusions or assumptions about new, unseen data exhibiting similar patterns. Developing a ML model to screen intubations for potential areas of improvement and to distinguish between attempts consisting of severe errors—where the trainee requires significant feedback/coaching—or common, simple-to-fix errors could thus improve efficiency of the training process by providing the educator with an indication of which attempts are of higher importance.

Local fire departments, training hospitals, medical education programs (e.g. ASTEC), and individual medical providers make up the anticipated customer base for a ML model for generating automated feedback for learners. Dr. Sami Sadeq, an experienced anesthesiologist working in the Advanced Surgical Technology and Education Center (ASTEC) at Inova hospital, mentioned there is hesitancy by medical centers and professionals to adopt AI technologies due to privacy concerns, but nonetheless, there is a strong interest in discovering how AI can improve patient outcomes and reduce healthcare professional’s workload.

The AQUARIUS framework, an AI model which combines automated detection and natural language processing to flag unusual images for review, is currently used in radiology [3]. Simulation training is another current solution for improving training efficiency and has been shown to improve trainee’s skills and the likelihood they form good habits [4]. However, these solutions do not directly apply to or replace case reviews with cadavers or live patients. Theator’s Surgical Intelligence Platform captures laparoscopic and robotic operating room videos and uses AI to highlight quality and training insights for hospitals [5]. Laparoscopy involves the surgeon creating a small incision into the patient’s abdomen and inserting a camera equipped to a telescopic rod to examine abdominal and pelvic organs for diagnoses or surgery [6]. This platform was not designed for ETI applications. Caresyntax provides video-based analytics within a data platform which records procedure video and audio to provide an analysis to enhance safety and procedure outcomes in the OR [7]. Caresyntax integrates patient health records, machine data, imaging studies, and surgical videos to identify opportunities for optimization. This platform does not provide feedback on video-assisted ETI procedures, but could be useful in combination with (external) ETI recordings for analysis of the patients' case within the OR. Additionally, Laerdal’s SimCapture captures and manages simulation videos and assessments using Objective Structured Clinical Examination (OSCE) workflows for education timeline management [8]. This platform, similarly to Caresyntax, focuses on integrating digital tools, such as trainee procedure recordings for manual revision, scheduling management, digital checklists, and scoring rubrics for a standardized training experience. Much like Caresyntax and Theator’s Surgical Intelligence, SimCapture does not provide feedback specific to ETI to be used by a range of medical providers, including but not limited to surgeons working within ORs.

By 2033 the global market size for AI in healthcare is projected to reach USD 505 billion growing at a compound annual growth rate of almost 40% over the next 6 years (2026-2033) [9]. Key factors for market growth and the goals with developing our model for video-assisted ETI are the same: improved efficiency, accuracy, and enhanced patient outcomes. The developed ML model could be increased efficiency, increased quality, and reduced cost for training on ETI. On

a broader scale, this would result in enhanced safety and patient outcomes; potentially reducing instances of patient trauma, hypoxia, or even death. When a patient suffers injury or death as a result of ETI errors, the provider may be subject to medical litigation and may be held liable through fines and license restriction or revocation. While training will not eliminate ETI errors, access to efficient, standardized, high-quality training resources, such as the proposed model and other AI solutions available on the market, will reduce the risk of fatal errors or medical negligence from occurring, making healthcare safer for all patients and training simpler for all healthcare professionals.

Overall, there is clear demand for AI in healthcare and a clear market opportunity for the model detailed hereinafter. With input from local healthcare providers and continued development by the authors, the developed model can be expanded to generate automated feedback for learners, such as coaching on ideal blade depth and tube navigation direction. The model described hereinafter successfully categorizes attempts as successful and unsuccessful, effectively indicating attempts which are the priority for receiving manual feedback.

Requirements/Specifications/Metrics/Standards

This design's primary requirement is to successfully classify video-assisted endotracheal intubation attempts as PASS or FAIL. A PASS classification indicates that the attempt met pre-established procedural and visual standards, while a FAIL classification indicates that the video should be prioritized for educator review and feedback. The system should process recorded GlideScope (or similar machinery) videos and assign a classification to each intubation attempt. The system must convert visual information from the recordings into a classifiable format from the original video data. The final design uses a supervised 2D convolutional neural network (CNN) and support vector machine (SVM). The 2D CNN extracts spatial features from video frames, while the SVM classifies the overall video as PASS or FAIL. The output must remain simple and interpretable so educators can understand the result without machine learning knowledge.

A key requirement is the model performs reliably with unseen videos—videos not used for training. To support this, the dataset was divided at the video level rather than the frame level. This prevents frames from the same video from appearing in both training and testing sets, which avoids artificially inflating performance metrics. Four-fold cross-validation was used so that each video was treated as unseen data in one fold. This provides more reliable performance results and gives a better estimate of how well the model may generalize to new videos.

The primary performance metric is accuracy, which measures the percentage of videos correctly classified, where a minimum of 80% was established as an acceptable accuracy. This threshold indicates that the system can support training review by correctly identifying most videos. In final testing, the 2D CNN + SVM model met this requirement across all tested resolutions, achieving accuracy between 83.38% and 83.81%.

The second key metric is the F1-score, which balances precision and recall. The F1-score is calculated as: $F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$. This metric is important because both false PASS and false FAIL predictions affect system usefulness, and the F1-score provides an indication of false positive and negative occurrences. A false PASS may allow a poor intubation attempt to receive inadequate attention, while a false FAIL could add unnecessary review burden for educators. The minimum acceptable F1-score for this project is 0.80. The final 2D CNN + SVM model achieved an F1-score of 0.843 across tested resolutions.

The model was also evaluated using the area under the receiver operating characteristic curve, or AUC. AUC measures how well the model distinguishes between PASS and FAIL videos across different classification thresholds. The receiver operating characteristic curve plots the true positive rate against the false positive rate, where: $TPR = \frac{TP}{TP + FN}$, $FPR = \frac{FP}{FP + TN}$.

The AUC can be represented as the area under this curve: $AUC = \int_0^1 TPR(FPR) d(FPR)$. An

AUC closer to 1.0 indicates strong PASS/FAIL separation. The final 2D CNN + SVM model achieved an AUC of 0.926, supporting its selection as the final design.

Training duration was also used as a practical performance measure. The 2D CNN + SVM model met this requirement, with training times ranging from 18 to 23 seconds across tested resolutions. In comparison, the 3D CNN required significantly longer training times, especially at higher resolutions, making this model configuration infeasible for real-world applications. Usability for educators is also an important design requirement. The current prototype generates either PASS or FAIL, which supports initial screening but future iterations of the model should provide anatomy-based feedback specific to each attempt to further support trainers and trainees. The model should be helpful, fast, and easy-to-use. However, it is important to note the system should not be used as a replacement for professionals, rather it is intended to be used as a “first-pass” review tool to help instructors determine which videos require further examination and instruction.

Another key requirement is secure and privacy-conscious data handling when evaluating intubations on real patients and cadavers. For this project, an AES-encrypted 2TB solid-state drive (SSD) was used for secure storage and transfer of video data, despite no real patient data used. The model was custom designed and stored locally to reduce the risk of unauthorized access and support responsible handling of medical training videos for potential applications using real cases (intubations) on patients.

Several standards and guidelines are relevant to this design. HIPAA Privacy and Security Rules apply because the project involves medical video data that may contain protected health information. These rules guide how identifiable information should be secured [10], [11]. NIST FIPS 197 is also relevant because it defines the Advanced Encryption Standard (AES), which supports the encryption of video data [12]. ISO/IEC 27001 provides a framework for managing information security, including access control, secure storage, and risk management for sensitive data [13]. Software, risk management, and usability standards are also important. IEC 62304 addresses the medical device software lifecycle from development through maintenance [14]. ISO 14971 supports identification of risks such as incorrect classification, data loss, or over-reliance on model output [15]. ANSI/AAMI SW96 provides guidance for managing security risks in medical device software [16]. IEC 62366-1 supports usability engineering, helping ensure that the system is user-friendly for trainers [17].

Because the final design uses supervised machine learning, ML-specific guidance was also considered. FDA Good Machine Learning Practice principles emphasize data quality, model evaluation, transparency, and documentation throughout the ML lifecycle [18]. Model Cards provide a structured method for reporting the model’s intended use, training data, performance, limitations, and appropriate use cases. These standards and guidelines help ensure that the design is accurate, secure, traceable, usable, and well-documented.

Final Design

Our current design consists of a Supervised 2D CNN + SVM combination, using spatial feature extraction from the 2D CNN and classification from the SVM to classify videos as PASS or FAIL. However, to better understand how this model works we need to dive into Convolutional Neural Networks (CNNs). CNNs are feedforward Artificial Neural Networks (ANNs) that are able to not only extract features from data, but also classify the information provided. CNNs are widely used in imaging and video classification as they are designed to process spatial or spatiotemporal data. This sort of ANN operates by applying multiple convolutional filters across an image for feature extraction. CNNs slide a kernel through an image layer by layer, compute the weighted sums of pixels to identify specific textures, edges, or patterns within the image which become features, to then compress these features into maps and feature vectors which indicate specific areas of an image that are relevant for diagnosis [19]. Lastly, by combining features from all layers, CNNs are capable of encoding simple and more complex patterns (the deeper the layers), allowing the model to assign different weights to specific features based on how important they are in classification. For these reasons, we believe that a CNN as the core feature extractor is the best choice. From existing literature, SVM classifiers are exceptional at identifying the distance between classes in an N-dimensional space, with N being the number of features being tracked. This particular classifier attempts to maximize the margin or space between the closest data points in opposing classes, allowing them to establish clear decision boundaries [20]. Furthermore, SVMs track as one of the most commonly used algorithms in video classification due to their high classification accuracy with small datasets.

After running a 4-fold Cross-Validation on 50 videos for training, the supervised 2D CNN + SVM hybrid model exceeded our initial expectations. The results showed a mean accuracy of 83.38%, a mean F1-score of 0.84, and a mean ROC-AUC of 0.93, setting the example as the best design for our application when compared to other preliminary designs such as the 2D CNN (alone) and 3D CNN. In addition to its fast training duration of less than 1 minute in comparison to up to 11 hours for the other CNN models, we believe this design model to be the most adequate lead towards the goal of this project. Conceptually, this model is still performing supervised classification; it learns from the labeled examples and assigns a class to unseen data. The overall structure and usage of this hybrid model is as follows: first, the 2D CNN spatial feature extraction looks at the individual frame and learns useful visual patterns; then it outputs a learned numeric representation of each frame, which is often referenced as an embedding or feature vector; finally, the learned features are passed into an SVM classifier, which performs the final classification, allowing for the predictions to be aggregated into a video-level PASS or FAIL output. However, this means our current model is limited by only being able to provide a binary classification and it lacks an understanding of each anatomical structure. An anatomical detector capable of integrating anatomical learning from the labeled structures (annotations) with the existing spatial information extracted by the 2D CNN would enable the model to provide detailed, practical feedback. The detector has been developed and

trained using an R-CNN with all 50 videos labeled with polygon annotations, but is still in the prototyping phase.

The team opted to train the detector with a Mask R-CNN model with a ResNet 50 + Feature Pyramid Network. Mask Region-based Convolutional Neural Network (Mask R-CNN) is a newer ground breaking algorithm proposed by Kaiming He et al. in 2017 [21]. Just like regular CNNs, R-CNNs are a sub family of CNNs that are specialized in object detection through region proposal and object classification. In reference to the original documentation by Meta researchers, Mask R-CNN adds a third branch in addition to the class label and bounding-box offset, known as the object mask, which is a much finer spatial layout of an object. This method is a great approach for our polygon labeled data as the model can learn both object localization and pixel-level masks in each frame. In our current implementation of this prototype, we decided to use the standard COCO-pretrained Mask R-CNN as is implemented for similar applications throughout existing literature. COCO, common objects in context, is a massive, widely-used dataset for computer vision and machine learning. We do not use the dataset itself, but we use the default COCO-trained Mask R-CNN weights, giving the detector a strong starting point for generic, simple features such as edges, shapes, and textures, but all training data comes from our LabelMe polygon annotations. The main reason for implementing this is to enable the model to learn labeled data (structures) in addition to the learned class data, as the goal is to detect and segment key anatomical structures (i.e. epiglottis, vocal cords, arytenoids, esophagus, and ET tube) to expand the capabilities of the model. Teaching the model the labeled data would be completed by tracking model outputs across each video and converting them into video-level anatomy features. These features would include several metrics such as how often each structure was visible and represent relative distances between the ET tube and key landmarks in addition to movement patterns. These features just like the 2D CNN + SVM hybrid model are classified through the same Mask R-CNN with a logistic regression classifier which predicts the final classification (PASS/FAIL) at the video level. Feedback parameters would be determined depending on the presence rate of these structures throughout the video. This section of feedback layer is still under work, but would be represented as the following parameters: airway alignment, representing positive feedback as the tube is near the vocal cords under a certain threshold, esophagus risk, the tube appears closer to the esophagus than to the vocal cords, uncertain alignment, an overall fallback when the anatomy is visible but there is not clear alignment present, possible alignment, representing when the tube was near the vocal cords at least once but does not exceed threshold, and limited anatomy evidence, which would represent feedback when the system did not observe enough reliable anatomy to make a strong interpretation under a threshold on vocal cords and endotracheal tube appearance rate.

Furthermore, to implement both of these models at the video decision level the pipeline takes the pass probability from the 2D CNN + SVM hybrid model and the pass probability from the anatomy detector to be combined with a weighted average. This sort of approach is known as ensemble learning, which is a machine learning technique that allows us to combine multiple individual models and produce a better overall production and a single model [22]. In our case,

this translates to an overall model that would learn complex spatial features from the 2D CNN + SVM hybrid model, in addition to object based features from the Mask R-CNN model in the anatomy detector. However, due to time constraints the anatomy detector has not been added into the final model, as more testing and further development is required. Results presented on the model herein represent the 2D CNN + SVM hybrid base model only.

Programming Environment & Implementation

The overall implementation of the model was done through Python 3.14.3 in Visual Studio. All packages were installed locally using Homebrew instead of .pip due to compatibility issues. The primary libraries used were PyTorch for designing and running the 2D CNN model and handling tensor computations; torchvision for image preprocessing and transformations applied to extracted frames; scikit-learn for implementation of the SVM classifier and evaluation metrics; and other common libraries such as NumPy, Pandas, and Matplotlib for numerical array manipulation, organizing results, and visualization of performance results.

We opted to deploy the model locally and avoid FastAPI or Azure, due to safety concerns regarding patient data on existing or cloud-based platforms. The script `ETI_classifier.py` was built based on the joblib files created from the trained 2D CNN + SVM model. Joblib files are files in Python that save or load large data objects. The folder in which the model was trained in contains the joblib files from each fold (0-3); here, based on the output from the joblib files, the script can vote whether the intubation attempt is a pass or fail. Through this deployment method we are also ensuring the reuse of the pipeline and pretrained ResNet-18 to extract visual features from the sampled video frames. We also ensure that the script is able to accept single or multiple .avi and .mp4 files, which the script takes in and breaks down into frames for the model to evaluate and aggregate. Regarding the final output, the prediction is printed out in the console, determining if the attempt was a pass or fail, and redirecting the user to the results folder created by the script, which contains more information regarding the probability of the video being a pass or fail as well as final decision, and number of frames in the video.

2D CNN + SVM Implementation

As mentioned before, the 2D CNN was responsible for transforming each frame into a high-dimensional feature representation, and instead of using its classification layer as well, the model extracts the intermediate feature vector produced by the network's final layer. In this feature vector, we can encounter complex spatial patterns that were learned from the image data, such as the structural shapes, textures, and spatial relationships within each frame. These extracted feature vectors then serve as the input to the SVM classifier, which identifies the best boundaries that separate the different classes (PASS or FAIL) [23]. More extensively, the classifier works by identifying the support vectors in the data, which are the training samples that are the closest to the decision boundary that is being created. Then the SVM optimizes this boundary with the goal of maximizing the margin between classes, improving the generalization

when predicting any unseen data [24]. Since our overall goal is to classify the entire video, the frame-level predictions were aggregated to generate a single video-level prediction. This means that by aggregating the frames, we combine the outputs into one final classification decision that represents the outcome of the intubation attempt. In Fig. 2, we can see a breakdown of the feature extraction and classification stages of the model. The figure includes a physical representation of how the 2D CNN feature extractor produces that feature vector, while the SVM classifier classifies the classes based on the feature vector.

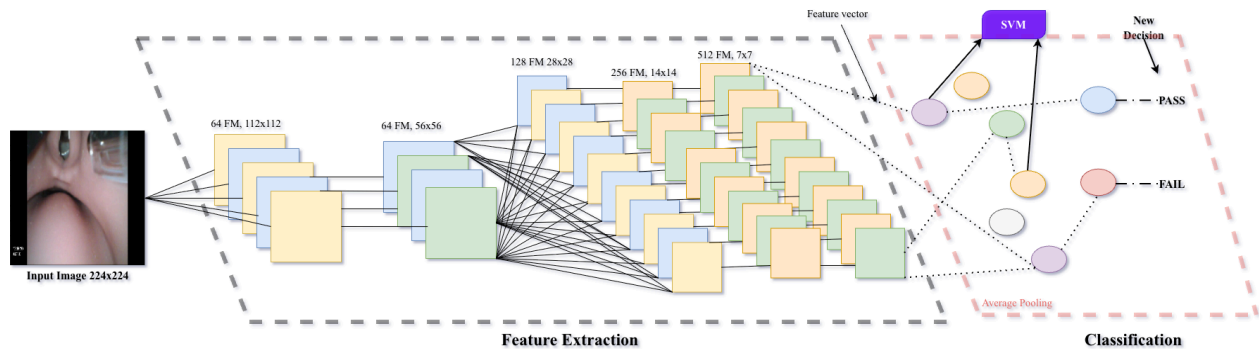


Figure 2: 2DCNN +SVM Classifier Pipeline Diagram

Overall, we believe that a 2D CNN + SVM hybrid architecture is the best configuration for our current design. This design has demonstrated stronger and more consistent performance despite the small dataset it was trained on. We believe that the combination of deep feature extraction of the 2D CNN and the classification nature of the SVM makes the system able to achieve a higher classification accuracy between these classes while maintaining a lower computational expense. Moreover, as previously mentioned, the implementation of an anatomy detector trained on 50 polygon-labeled videos by a Mask R-CNN model would provide a layer of feedback for the user in comparison to the single interpretation from our 2D CNN + SVM. Future implementation should focus on prototyping and implementation via ensemble learning which would fuse the hybrid visual classifier based on ResNet-18 embeddings and a SVM model, with an anatomy aware model based on Mask R-CNN detections, being able to track and detect anatomical features at video-level, providing more detail and non-binary feedback to the user.

Verification & Validation

General Testing Methodology & Programming Environment

The dataset used was recorded at the Advanced Surgical Technology and Education Center (ASTEC) at Inova Fairfax from a male manikin. A total of 53 intubation videos were recorded, but 3 were excluded from model training due to extreme differences in how the procedure was performed. One excluded video was recorded on a different manikin with the hopes of adding variability to our data, but the manikin lacked accurate anatomy. The final dataset used consisted of 50 videos, where 25 were PASS and 25 were FAIL. To prototype all three model designs and choose the best-performing one, we initially only utilized 20 for training and testing due to time constraints and the high computational expense of training CNN models. Due to the small dataset able to be used for model training, a 4-fold cross-validation was selected, meaning the set is divided into 4 subsets, where one subset was held aside for testing while the others were used for training. The process is repeated across all folds so each video is evaluated as “unseen” test data. Fold cross-validation is a robust tool that aids in obtaining more robust and unbiased estimations of the ML’s performance. By using this method we reduce variance in performance metrics and prevent overfitting better than by using a single train-test split. The two annotation conditions we evaluated during model development were rectangles and polygons. These annotations help us define our anatomical regions of interest that we want the model to learn and identify. Within these tests, the model(s) were evaluated under both labeling conditions to determine how the geometry of each annotation style impacts classification and performance. Additionally, both the 2D CNN and 3D CNN models were trained with a symmetrical lower resolution of 128x128 for quick prototyping and statistical analysis, while the 2DCNN + SVM model was trained with a resolution of 224x224. Moreover, to ensure a fair model comparison, we decided to split the videos at the video level, therefore creating partitions for the training and testing sets. This helped us prevent frames from both the training and testing sets from appearing in the same video set, therefore avoiding artificially inflating model performance.

To determine model performance, we utilized common statistics used in all machine learning (ML) applications, such as accuracy, precision, recall (sensitivity), specificity, and F1-score. All of these metrics give us a comprehensive evaluation of how the models classify the data and help us identify the trade-offs between performance and false positive or false negative rates. Additionally, post-statistical analysis, such as the mean performance and standard deviation across folds, was reported between models across all folds to assess the consistency and reliability of the models. Such a statistical approach was also taken into account when examining the presence rate of anatomical structures that were labeled to assess how each label type affected the model’s recognition of each structure.

Test 1: Tracker Shape Test

Two tracker shapes, polygons and rectangles, were evaluated based on their influence on model performance. The sample size, the number of videos used to train and evaluate the model, is limited to a total of 20 videos for each tracker shape, with an even split of ‘PASS’ and ‘FAIL’ intubation attempts. All attempts were performed by Dr. Sadeq on an adult, male manikin using a GlideScope machine from ASTEC. Dr. Sadeq was instructed to perform good and bad intubation attempts, but was not provided any additional instructions. To assess model performance with the different tracker shapes, each model type was trained using one set of labeled videos, tested with non-labeled videos, and evaluated using accuracy.

To set up the experiment, LabelMe was used to label key structures using both tracker shapes, or annotation conditions, in each frame for all videos within the dataset. Key structures include the esophagus, arytenoids, vocal cords, and endotracheal tube. Fig. 3 and 4 are sample frames labeled with rectangles and polygons respectively.

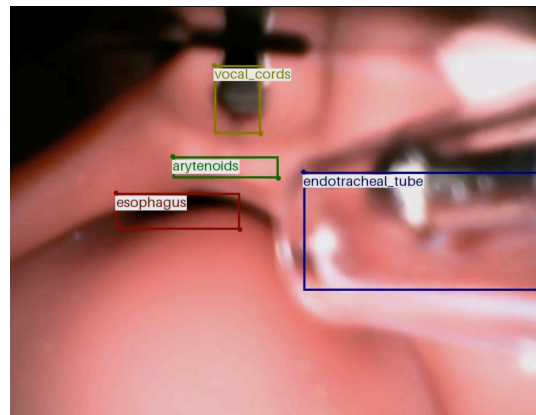


Figure 3. Frame with labeled rectangles on anatomical structures and displayed labels

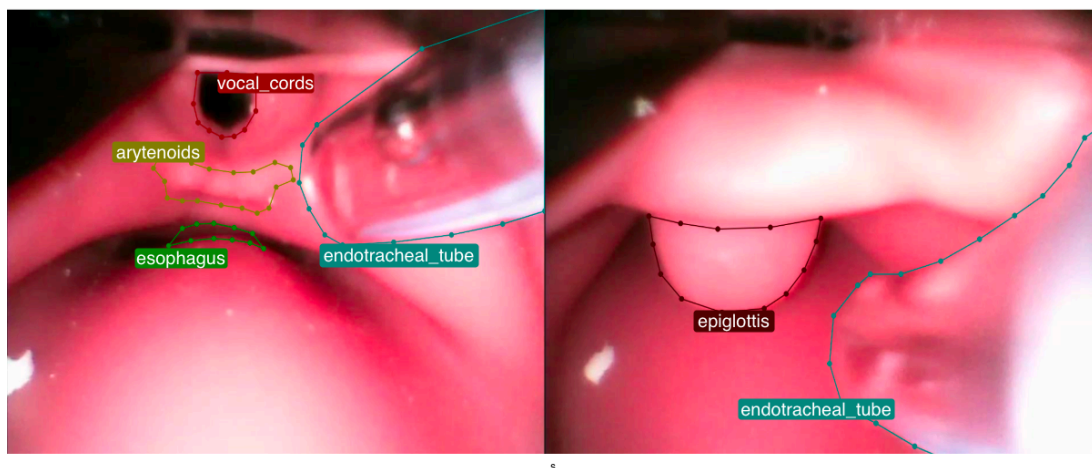


Figure 4. Frames with labeled polygons on anatomical structures and displayed labels

Two types of polygon labels were used during the labeling process: AI-generated polygons and manually-created polygons. AI-polygon trackers proved to be too computationally demanding for personal laptops; some members were required to use only manual polygons. Manually-created polygons proved to be more efficient and easier to use due to the added control over the number of boundary points, as opposed to a random number of points (~30-150) from the AI-polygon tool. The majority of the videos were, therefore, labeled using manually-created polygons. Labeling was completed over the span of a few weeks. The test was then run on all potential model designs (i.e. 2D CNN, 3D CNN, and SVM).

We used a 4-fold cross-validation, and for each fold, we trained the model on the training split and saved the best-performing model checkpoint for that fold. After training, we used a dedicated evaluation program named `eval.py`. This script automatically loads the saved model for each fold, runs the model on the fold’s held-out test videos, and produces the plots and metric tables used for reporting. During evaluation, each video was split into multiple short clips (default: 5 per video) to stabilize predictions. The model produced a score for each clip, which is combined into a single score per video by averaging them. The video score or mean logit is converted into a probability using the sigmoid function, then it is classified using a fixed threshold of 0.5 (probability $\geq 0.5 \rightarrow$ PASS; otherwise, FAIL). Using the predicted labels and ground-truth labels, `eval.py` generated the fold-level accuracy.

The annotation evaluation script evaluates whether the model performs differently depending on which anatomical structures are visible. To do this, it reads the per-video prediction results produced by `eval.py` (true label vs predicted label for each video) and checks the LabelMe annotated JSON files for that video to determine whether an anatomical structure appeared in the labeled frames. A presence rate is calculated for each annotation (fraction of frames in a video containing that label), then reports performance summaries including: presence rate, accuracy when the anatomical structure is present, accuracy when the annotation is absent/unknown, the accuracy difference (present minus absent/unknown), and the misclassification rate when present. These summaries were exported as CSV tables and helped identify which visibility conditions are most strongly associated with correct vs incorrect model predictions [25].

The test was successful as we were able to identify a preferred annotation type for labelling of the remaining videos in our dataset. The results show the polygon labels outperform the rectangle annotations across the tested model designs, suggesting the polygon labels better define anatomical regions and spatial relationships between tracked/labeled structures for our ML application. See test results below in Table 1.

Polygon	3D CNN	2D CNN + SVM	2D CNN
Accuracy	0.85	0.85	0.75
F1 score	0.88	0.88	0.60

Rectangle	3D CNN	2D CNN + SVM	2D CNN
Accuracy	0.70	0.90	0.50
F1 score	0.76	0.92	0.39

Table 1. Summary of model accuracy for both polygon and rectangle annotations with sample size (total No. of videos used for model training and testing) = 20.

Test 2: Model Design Comparison Test

The main goal of this test was to compare our distinct ML designs: 3D CNN and 2D CNN in combination with SVM. This was done in order to determine the optimal model for correctly classifying video-assisted ETI attempts. While we did originally test a third model (2D CNN) with a smaller dataset of 20 videos (see Table 1), the model performed poorly and was eliminated from further testing.

To provide an in-depth evaluation and mitigate high variance in preliminary testing, the dataset was increased to include 50 labeled videos. Based on the results of Test 1, the remaining 30 videos were fully annotated exclusively using the polygon tracker tool. This dataset was used for model training from here out. The two remaining model designs were trained and tested using these videos, and the validation was conducted utilizing a 4-fold cross-validation strategy. For each iteration, the models were trained on three folds and tested on one completely unseen fold. This process was then repeated four times, allowing for every video to be a part of the unseen test class exactly once and ensuring that the models were being evaluated on their ability to generalize rather than just memorize data.

To prevent overfitting and maintain consistency, some general parameters were standardized across all of the classifying CNNs. This included standards like capping training at 20 epochs once it became clear that further training only increased computing time without significantly improving the accuracy of the models. Additionally, the ADAM (Adaptive Moment Estimation) optimizer from the Torch library was utilized to update the models' parameters based on the computed gradients during training. Those parameters were then used to compute individual adaptive learning rates with a base learning rate set to 0.004.

Outside of the general parameters, each ML model design was constructed and processed slightly differently. To start, the 3D CNN model was designed to learn both spatial and temporal patterns simultaneously by using a sequence of frames as its input. This breakdown of video clips allows the model to capture the motion and trajectory of the endotracheal tube. However, to offset the high computational cost of 3D CNNs, the input frames were resized to a ratio of 128x128. Unlike the 3D CNN model, the combination model design used both 2D CNN and SVM to separate feature extraction from the final classification. This means that the 2D CNN was used exclusively to generate deep visual embeddings for each (resized 224x224) frame. These high-dimensional features were then averaged per video and fed into the SVM portion of the model utilizing a linear kernel in order to actually classify the video. This hybrid approach leverages the CNN's ability to learn visual structures and the SVM's mathematical efficiency in establishing clean decision boundaries for binary video classification.

Test 3: Resolution and Duration Test

Image resolution plays a significant part in the training and testing of any deep learning model. However, in most medical applications, resolution is sometimes sacrificed to adhere to a fast computational model, leading to the elimination of details in these images that would otherwise be important [26]. In this test we aim to evaluate how both models, the 3D CNN and 2D CNN + SVM, handle training and testing on videos of various input resolutions in order to determine which model design is the most efficient and identify the lowest effective resolution.

Both models were trained and tested with the videos at different resolutions (64x64, 128x128, 224x224, 320x320, 600x600) based on the paper by Carl et al. [27]. The results of each model's run at the various resolutions were compared using AUC, model accuracy, and F1-score.

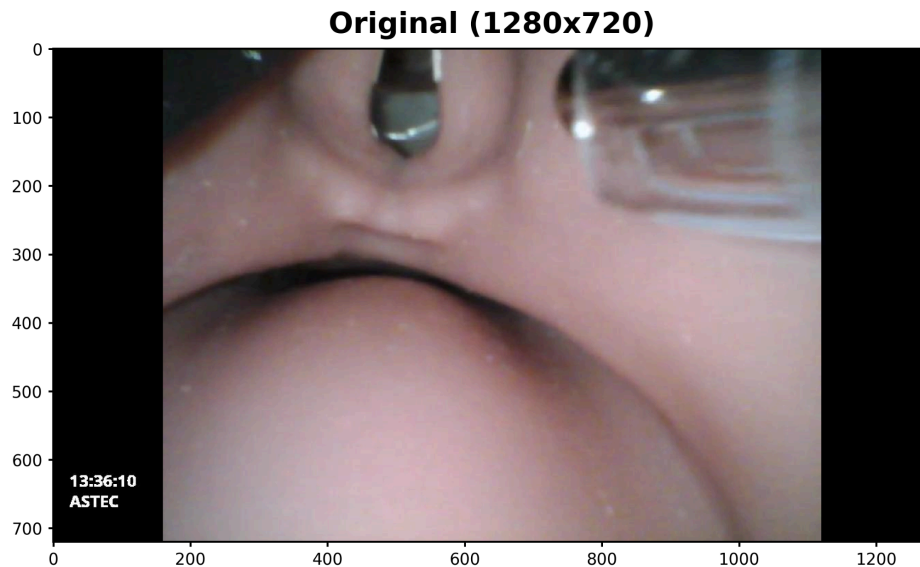


Figure 5: Original Image Resolution

The original resolution of each video in the dataset is 1280x720, shown in Figure 5, in an .avi format, and the videos were then converted into a symmetrical resolution for processing by the models. Furthermore, as part of preprocessing, the videos were broken down into frames using the video-cli library, which allowed us to utilize the frames for labeling, training, and testing the models. An example of how one of these frames may look at various resolutions can be seen in Figure 6 below.

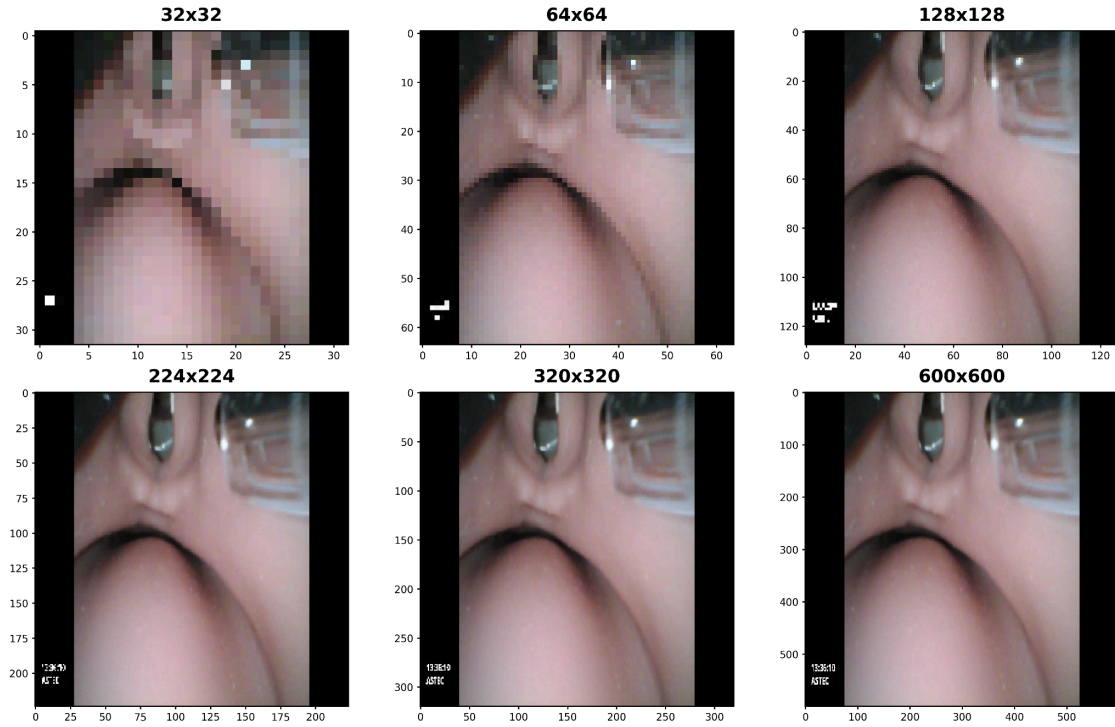


Figure 6: Aspect Ratio Comparison from the Original Image (Figure 1)

In addition to comparing standard model parameters (e.g., accuracy, AUC, etc.), we also aimed to compare the training and testing duration of the models. The duration parameter was added as long duration time for model training and evaluation would make the model an unsuitable tool for expediting feedback and reducing workload.

Test 4: Clinician and Model Comparison Test

Six experienced healthcare providers, or clinicians, were asked to watch 10 videos (5 PASS, 5 FAIL) and answer two questions about each. First, they were asked to select the option that best describes the procedure performed. Then, depending on the option selected in the first question, the clinicians are asked to provide feedback on the video. Figure 7 shows the response page for video 1. The link towards the top of the page directs responders to the respective video in .mp4 format for universal playback compatibility. The form was created on Google Forms, and the videos are hosted in a Google Drive folder.

The screenshot shows a Google Form titled "Intubation Feedback Model Evaluation". Below the title is a blue header for "Video 1 - '1'". A link to a Google Drive video is provided: https://drive.google.com/file/d/1z7y6fOWKI9x3ck9d8HjEL62GkfCae-jp/view?usp=drive_link. The main question asks: "Select the option which best describes the procedure performed in video 1." with three radio button options: "Good attempt (PASS). No feedback.", "Good attempt (PASS). Requires feedback.", and "Bad attempt (FAIL). Requires feedback.". Below this is a text input field labeled "Please provide feedback for video 1." with the placeholder "Your answer". At the bottom are "Back", "Next", and "Clear form" buttons.

Figure 7. Healthcare provider evaluation form, video 1.

Clinicians were asked to join the authors for a brief 10-15 minutes to describe our project (if we had not previously met with them) and provide additional instruction regarding the evaluation form. An example of additional instruction provided to the responders was to provide feedback with the assumption that the learner is familiar with anatomy and ETI procedure steps (i.e., medical jargon is acceptable).

Responders generally agreed on PASS/FAIL descriptions and provided similar feedback on videos requiring it. Videos 6 and 7 were the only exceptions, as a majority of responders considered these attempts to be passing while 1-2 considered them to be failed attempts (see Fig. 8). The majority response was considered the “clinician response” for analysis purposes.

Select the option which best describes the procedure performed in video 6.	Please provide feedback for video 6.	Select the option which best describes the procedure performed in video 7.	Please provide feedback for video 7.
Bad attempt (FAIL). Requires feedback.	The ETT was pulled out after being correctly inserted. The blade needs to be midline.	Bad attempt (FAIL). Requires feedback.	ETT dislodged, blade not midline and inserted too deeply
Good attempt (PASS). Requires feedback.	Likely successful, but unable to visualize tube passage due to poor technique	Good attempt (PASS). Requires feedback.	Off axis, poor engagement of hyoepiglottic ligament
Good attempt (PASS). Requires feedback.	Off-axis view prevents visualization of glottic opening and ETT placement. Although successful, this orientation will likely result in future missed attempts and does not record the correct placement of the ETT.	Good attempt (PASS). Requires feedback.	Off axis camera view and blade insertion too deeply provide visual obstructions and make ETT placement difficult.
Good attempt (PASS). Requires feedback.	Off axis to right. Poor visualization of cords	Good attempt (PASS). Requires feedback.	Initially tube headed toward esophagus, but backed out and successfully corrected. Off axis to right. Poor visualization of cords.
Bad attempt (FAIL). Requires feedback.	Visualization not optimal, tube placed correctly but then pulled out.	Good attempt (PASS). Requires feedback.	Endotracheal tube placed correctly, took a bit of time, other structures hit prior to placing the tube, and proper visualization was not achieved.
Good attempt (PASS). Requires feedback.	Good attempt. Always lead with suction, SALAD method. Next time, try approaching the airway from straight on. The laryngoscope should be midline in the patient's mouth. This will achieve a better angle and view to insert the rigid stylet. Always wait till the cuff is inflated and the rigid stylet is removed before removing the laryngoscope.	Good attempt (PASS). Requires feedback.	Good attempt. Remember to always lead with suction, SALAD method. Next time try keeping the laryngoscope midline and straight. This will help improve the correct blade placement and view of the vocal cords.

Figure 8. Clinician responses for videos 6 and 7.

The clinicians unanimously agreed with the true label and model output for 8 of 10 videos. "True label" refers to the intended or "correct" classification by group members and Dr. Sami Sadeq, who is performing each intubation. For videos 6 and 7, the clinicians in the minority agreed with the true label and model output, but the majority did not. For videos 2 and 3, the clinicians agreed with the true label (both FAIL), but the model did not. The model therefore failed for these two cases.

From this test alone it is difficult to conclude model accuracy compared to clinicians due to the various interpretations of what constitutes a good (PASS) or bad (FAIL) attempt. However, the group does not feel it appropriate to draw the distinction, as no member is currently trained or under training to work as a healthcare professional. The model showed an accuracy of 80% compared to true labels but an accuracy of only 60% compared to clinicians alone. Additional feedback from other clinicians and a clearer, common distinction between attempt categories would improve the validity of this test.

Discussion

Test 1

In this initial phase of testing, the intention was to evaluate the influence of the geometry of the annotation tracking shapes (e.g., polygon vs. rectangular) on the models' overall classification performance as well as spatial feature extraction capabilities. As demonstrated in Table 1, there was a relatively distinct impact on model performance based on which tracking label shape was utilized. However, primarily, this test revealed the major limitations of the standard 2D CNN model. This model design failed to meet the minimum 0.80 acceptance criteria across all parameters, only managing an accuracy of 0.75 and an F1-score of 0.60 for the polygon-annotated videos and achieving even lower scores for the rectangular-annotated videos. The most likely cause of this failure is that the standard 2D CNN lacked the mathematical mechanisms needed to effectively average individual 2D frame predictions into a cohesive decision regarding a complete video. With this in mind, the team moved to exclude the 2D CNN model design from further testing and instead focus on the 3D CNN and 2D CNN + SVM hybrid models.

Another revelation revealed during this test was that the rectangular labels provided a slight boost in performance by the 2D CNN + SVM hybrid model when making the PASS/FAIL classification. With this combination, the model achieved a 0.90 accuracy and 0.92 F1-score compared to the 0.85 accuracy and 0.88 F1-score achieved with polygons. This jump in model performance is likely a result of the rectangular bounding boxes naturally capturing more background context around the structures, helping the SVM to establish boundaries that allow it to better separate successfully performed intubations from failed ones. However, a simple binary classification is only part of the final goal in terms of the model's output.

In order to develop the model into a comprehensive training tool, an R-CNN anatomy detector is currently being prototyped to be used simultaneously with the binary classification portion of the model and provide more detailed feedback to users. This anatomy detector would be able to recognize, isolate, and track specific anatomical structures within the airway, and by doing so, it can provide information regarding the decision-making behind the video classification, such as "ET tube visible in 40.3% of indexed frames," "vocal cords visible but tube does not clearly approach them," and "esophagus visible in 65.7% of indexed frames." Nevertheless, prior to building this detector, it was necessary to establish how accurately each tracker shape allowed the models to recognize and isolate specific anatomy. The supplementary data in the table below, gathered during Test 1, was crucial to make an informed decision regarding the development of the detector as well as to make a final decision regarding which tracker shape to move forward with in future testing.

Accuracy when Present (Polygon)			
Label Name	3D CNN	2D CNN + SVM	2D CNN
Vocal Cords	88%	88%	69%
Epiglottis	83%	100%	83%
Arytenoids	89%	89%	72%
Esophagus	89%	89%	72%
Endotracheal Tube	85%	85%	75%
Accuracy when Present (Rectangle)			
Label Name	3D CNN	2D CNN + SVM	2D CNN
Vocal Cords	67%	83%	50%
Epiglottis	80%	83%	50%
Arytenoids	67%	83%	50%
Esophagus	67%	83%	50%
Endotracheal Tube	70%	75%	50%

Table 2. Summary of label accuracy for both polygons and rectangles.

The data in Table 2 clearly shows the superiority of polygon labels when isolating specific anatomical structures. For example, when the epiglottis was present in frames labeled with polygons, the models achieved an average accuracy of 88.67%, whereas rectangular labels yielded a noticeably lower 71%. This makes polygon labels ideal for the purposes of the anatomy detector because of the polygon annotations' ability to tightly conform to the specific contours of the airway and prevent excess background context from harming the model's ability to recognize features. This precision of the polygon labels and the importance of including the anatomy detector to provide detailed feedback make the slight decrease in binary accuracy worthwhile and are why polygons were selected as the final annotation method for the full 51-video dataset.

It is also worth noting that because all of the ETI procedures we recorded were performed on a single manikin, by the same clinician, there is only so much diversity in the dataset. This has the potential to be a major limitation in this model's design, as it is unlikely to be able to generalize to videos performed on different kinds of manikins (e.g., pediatric, different models) or real patients. This limitation means that, in the future, it is incredibly important to obtain a larger, more diverse dataset that will allow the model to learn all possible scenarios that may be uploaded by a user.

Test 2

Following the decision to exclude the standard 2D CNN and settle with polygon labeling, the testing phase shifted to a more rigorous comparison between the 3D CNN and 2D CNN + SVM hybrid model designs. As seen in Table 1, preliminary testing with a smaller dataset (only 20 videos total) revealed that both models were performing well above the 0.80 acceptance criteria. However, once the dataset was expanded to the full 50 videos, the overall performance of both models actually dropped.

	Acc.	F1 Score	Precision	Recall
3D CNN	0.615	0.739	0.599	1.00
2D CNN + SVM	0.838	0.844	0.794	0.917

Table 3. Summary of each model's performance with polygon labels.

This drop in performance is illustrated in the data above, and it can be seen that while the 2D CNN + SVM model still performs relatively well, and most parameters meet the 0.80 or better criteria, the 3D CNN significantly suffered from the increased dataset size. The drop in performance across both models could also be attributed to all the attempts being recorded on the same manikin in the same environment. This severe lack of diversity creates a high risk of the model overfitting to specific techniques and simulating anatomy. Additionally, the drop in accuracy highlights the idea that scaling up a relatively homogeneous dataset can actually compound errors rather than improve learning.

It is worth noting that it was unexpected for the 3D CNN to decrease in performance so severely. The team initially hypothesized that the 3D CNN would be the superior design of the original three options. This assumption was made because 3D CNNs are designed to learn both spatial and temporal patterns simultaneously by using a sequence of frames, and it was expected to accurately capture the motion and trajectory of the endotracheal tube over time. However, expanding the dataset revealed that this temporal sensitivity actually became a liability. While the 3D CNN achieved a perfect recall of 1.00, this metric is highly misleading when the low precision of 0.599 is taken into account. A perfect recall occurring at the same time as a low precision score indicates the 3D CNN overcompensated by predicting a single class for nearly every video. This means it failed to accurately distinguish between the distinct features that affect the classification of a Pass versus a Fail attempt, leading to a high rate of false positives, an F1-score of 0.739, and an overall accuracy of just 0.615.

On the other hand, the 2D CNN + SVM hybrid model maintained a balanced performance profile when the dataset was increased, achieving an accuracy of 0.838 and an F1-Score of 0.844. Most importantly, the hybrid model achieved a precision of 0.794 and a recall of 0.917. This balance proves that the model is actively learning and distinguishing the correct spatial features rather than guessing a single class. The SVM's ability to establish clean

mathematical decision boundaries based on the spatial visual features extracted by the 2D CNN proved highly adaptable to the variance of the 50-video dataset, proving it to be the superior machine learning design for this clinical application.

Test 3

To further validate the model comparison in Test 2 and ensure the final design was able to balance processing speed and visual fidelity, a supplementary resolution and duration test was conducted to evaluate both models across various input resolutions (e.g., 64x64, 128x128, 224x224, 320x320, and 600x600).

Image Size	Metric	3D-CNN	2D-CNN + SVM	Improvement
64x64	Accuracy	50.00%	83.81%	+33.81%
	Duration	18m 36s	18.19s	~61x faster
	AUC	0.648	0.926	+42.90%
	F1-Score	0.666	0.843	+26.58%
128x128	Accuracy	61.11%	83.38%	+22.27%
	Duration	35m 26s	21.31s	~100x faster
	AUC	0.833	0.926	+11.16%
	F1-Score	0.711	0.843	+18.57%
224x224	Accuracy	61.54%	83.81%	+22.27%
	Duration	1h 28m 02s	23.15s	~228x faster
	AUC	0.747	0.926	+23.96%
	F1-Score	0.739	0.843	+14.07%
320x320	Accuracy	53.85%	83.38%	+29.53%
	Duration	2h 53m 46s	21.32s	~489x faster
	AUC	0.675	0.926	+37.19%
	F1-Score	0.685	0.843	+23.07%
600x600	Accuracy	53.85%	83.38%	+29.53%
	Duration	10h 47m 28s	21.28s	~1,825x faster
	AUC	0.623	0.926	+48.64%
	F1-Score	0.684	0.843	+23.25%

Table 4. Improvement comparison between 3D-CNN & hybrid model.

The resolution testing consistently reaffirmed the idea that the hybrid 2D CNN + SVM model was the superior model design. As shown in Table 4 above, the 2D CNN + SVM model consistently outperformed the 3D CNN across accuracy, AUC, and F1-Score at every tested resolution. On the other hand, the 3D CNN's metrics fluctuated but remained comparatively low, only peaking at 61.54% accuracy at 224x224.

Beyond the usual assessment parameters (e.g. Accuracy, F1-Score, and AUC), this test highlights a critical operational constraint, efficiency.

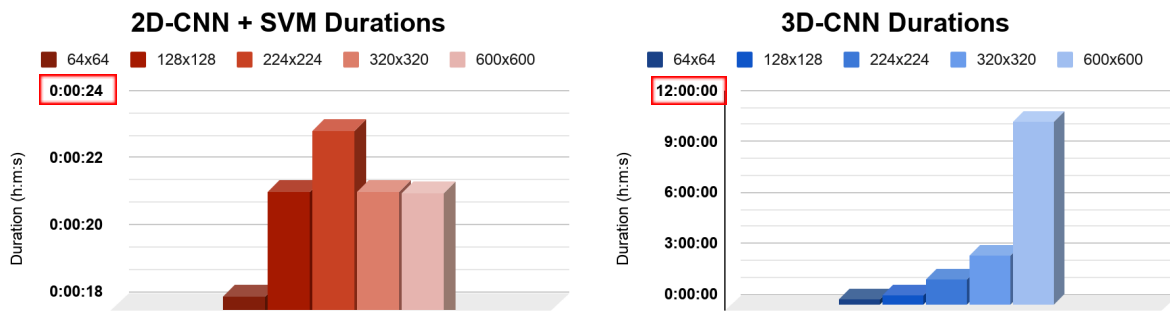


Figure 9. Training duration comparison between models

The 3D CNN model struggled with the weight of processing tons of high-resolution data simultaneously. So, as image sizes increased, the processing time required to run the 3D CNN increased exponentially, starting at roughly 18.5 minutes at 64x64 and taking nearly 11 hours (10h 47m 28s) at 600x600. This renders it impractical for use as a rapid clinical feedback tool. In contrast, the hybrid model maintained a consistently rapid processing duration that averaged between 18 and 23 seconds per run regardless of image resolution. This means that the model design that provides the most efficient trade-off between accuracy and efficiency was the 2D CNN + SVM hybrid model utilizing an input resolution of either 64x64 or 224x224. For the purposes of our model, it was determined that a 64x64 resolution was too low to be considered fully reliable, so 224x224 was decided as the standard input resolution.

Test 4

The final measure of the model's success is its alignment with actual medical professionals. The clinician and model comparison test was designed with the objectives to first determine if the hybrid model's automated PASS/FAIL classifications mirror the judgment of experts and second, to collect qualitative clinical responses to inform the type of detailed feedback being output by the upcoming R-CNN anatomy detector.

The evaluation currently has a very limited sample size, with only six clinician responses having been received. Nevertheless, as previously mentioned, 1-2 clinicians provided directly conflicting categorizations for Videos 6 and 7, highlighting them as complex cases. One clinician

assessed both procedures as successful "good attempts" (PASS), while another rated both as "bad attempts" (FAIL). For Video 6, one of the failing evaluations noted that the endotracheal tube was pulled out after insertion and the blade was not midline. Conversely, the passing evaluation noted poor visualization technique but deemed the attempt "likely successful" regardless. Similarly, for Video 7, the failing evaluation once again cited a dislodged endotracheal tube as well as an overly deep insertion, while the passing evaluation cited poor engagement of the hyoepiglottic ligament. This divergence in expert opinion underscores the necessity of a standardized, automated evaluation tool, as human reviewers can clearly disagree on procedural nuances.

In regard to the model comparison with clinician responses, the results of which can be seen in Table 5 below, the model failed to correctly classify a video in accordance with both the true label and the clinician evaluation on two occasions (Videos 2 and 3).

Video ID	True Label	Model Predicted Label	Clinician Evaluation (Majority)	Disagreement (How many?)
1	PASS	PASS	PASS	no
10	FAIL	PASS	FAIL	no
13	FAIL	PASS	FAIL	no
18	PASS	PASS	PASS	no
25	FAIL	FAIL	FAIL	no
28	FAIL	FAIL	PASS	yes, 2
35	FAIL	FAIL	PASS	yes, 1
41	PASS	PASS	PASS	no
51	PASS	PASS	PASS	no
54	PASS	PASS	PASS	no

Table 5. Summary of clinician responses in comparison to model output.

Additionally, for the complex videos, 6 and 7, the model correctly classified the videos as failures based on the true label but technically did not match the majority consensus of clinicians. This discrepancy once again showcases how subjective manual review can be, as the true labels were based on the opinion of the medical professional performing the procedure. It is also worth noting, with a larger response pool, the final evaluation of these two videos could potentially change.

Summary & Conclusion

In contrast to existing clinician-based methods for reviewing endotracheal intubation videos, this prototype utilizes a machine learning algorithm to expedite and streamline the review process. It assists instructors in identifying which trainee video recordings were submitted for initial review and require further consideration. The technology is designed to support, rather than replace, clinician judgment by facilitating the selection of optimal videos for educational purposes based on preliminary automatic PASS or FAIL classifications.

A two-dimensional convolutional neural network (CNN) combined with a support vector machine (SVM) classifier was selected after evaluating multiple models using accuracy, F1 score, area under the receiver operating characteristic (ROC) curve (AUC), and processing time as performance metrics. The standard 2D CNN did not achieve satisfactory results in pilot tests, and the 3D CNN exhibited increased processing time and performance variability. In contrast, the 2D CNN with SVM classifier demonstrated superior performance and efficiency. The 2D CNN and SVM model met the key performance requirements, achieving accuracies of 83.38%, 83.4%, and 83.81% across different resolutions, all exceeding the 80% target. The model achieved an F1 score of 0.843 and an AUC of 0.926, indicating strong separation between PASS and FAIL classes. Training time ranged from 18 to 23 seconds, providing a faster review process than the 3D CNN model. A resolution test was also conducted on both model designs to evaluate model performance at low and high resolutions, as well as provide an indication of training time required at each resolution. The 3D CNN total training time went up to almost 12 hours for the highest resolution (i.e., 600x600), whereas the time was drastically cut down to almost 22 seconds at the same resolution for the 2D CNN/SVM hybrid. Model performance was similar across resolutions for each model type. The hybrid configuration had superior accuracy, F1-score, and AUC overall compared to the 3D CNN model at different resolutions. The 64x64 and 224x224 resolutions for the hybrid model performed the same, but the higher resolution was selected for further development due to it being most commonly used in deep learning applications. Additionally, the higher resolution is expected to perform better long term with higher quality data.

The project assessed annotation methods to guide future development. Rectangle labels were generated faster and yielded satisfactory accuracy for binary classification. However, polygon labels enabled more precise identification of anatomical structures, including the vocal cords, arytenoids, esophagus, and endotracheal tube. To support the long-term objective of delivering detailed feedback beyond PASS/FAIL, the use of polygon labels is recommended for future anatomical detector development.

Clinician validation commenced through Google Form evaluations of selected videos. Initial findings indicated general consensus among clinicians regarding most classifications, with occasional disagreements. These results highlight the subjectivity inherent in endotracheal intubation (ETI) review, underscoring the need for a standardized first-pass review tool. Full

model validation will occur once additional clinician responses are collected and a consensus is established.

In summary, the project demonstrated that machine learning can classify simulated video-assisted intubation attempts with promising accuracy and efficiency. Key limitations include a small dataset, reliance on a single manikin, limited clinician responses, and binary output only. Future research should expand the dataset, include more diverse users and recording conditions, gather additional clinician evaluations, and finalize the Mask R-CNN anatomy detector. Integrating anatomy-based feedback would allow the system to explain PASS or FAIL outcomes, thereby increasing its value for intubation training.

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Appendix

Appendix A. Budget

The project budget focused on secure data storage, annotation software, and final presentation materials. The team did not purchase a backup encrypted SSD, so it was removed from the final budget. LabelMe was paid out-of-pocket by the team and cost \$80.

Item	Quantity	Cost	Purpose
2TB AES-encrypted SSD	1	\$199.99	Secure storage and transfer of de-identified video data
LabelMe	1	\$80.00	Video frame annotation using rectangle and polygon labels
Poster printing / final materials	1	~\$50.00	Final poster and presentation materials

Table A1. Final project budget for BENG 492 and BENG 493.

The estimated final project cost was approximately **\$330**. The final cost was lower than the original estimate because the team did not purchase a backup encrypted SSD or additional supporting hardware. The main expenses were the encrypted SSD, LabelMe annotation software, and final poster materials.

Appendix B. Member roles & responsibilities

Each member plays a role in the completion of course deliverables and project development. In the fall, members completed a team charter where we initially developed a table of general roles and responsibilities to be held by each member. Table B1 shows an updated version of the original table created in September 2025.

<u>Role</u>	<u>Group Member</u>	<u>Responsibilities</u>
Project Manager	Ace	*Main point-of-contact between team members *Establish meeting agendas & group deadlines *Ensure team meets deadlines & completes tasks *Biological background and research
Machine Learning & Programming Officer	Nico	*ML model background research & implementation. *ML model prototype testing and implementation.
Research Officer & Support	Sedef	*Biological background and research *Locating additional intubation images/videos
Financial & Communications Officer	Quynh	*Main point-of-communication with client, dept coordinator (Amal Nadel), & other external contacts *Estimates cost & orders material through dept
Test & Evaluation, Project Delivery	All Team Members	*Development of prototyping (testing) procedures *Deliverable preparation (i.e. preparation of presentation slides and scripts)

Table B1. Roles and responsibilities of each member per the Fall 2025 Team Charter

Since the completion of the team charter in September, tasks have been assigned to each member for key project milestones and deliverables by the project manager. An extensive list of member responsibilities and tasks is shown in Table B2.

<u>Group Member</u>	<u>Responsibilities & Tasks</u>
Ace	*Assign all tasks/responsibilities to group members. *Primarily led internal group conversations and communication with Dr. Ross.

	* Scheduled team meetings, maintained meeting minutes & established meeting agendas. *Created & maintained Weekly Work log. *Wrote design history file organization document & maintained file organization. *Wrote interview questions for background & market research assignment, requirements section of design requirements & standards assignment. *PDR presentation & report: background & relevance; need statement; gantt chart; alternative design (SVM). *CDR presentation & report: problem & need statements; gantt chart; introduction; alternate design (SVM). *HW1 (493): test 1 (tracker shapes) *HW2 (493): test 1 (tracker shapes) *Labelled videos with polygon tracker: 2, 4,5, 6, 20, 25, 32, 33, 37, 39, 40, 41, 43, 44, 45, 52 *Mid semester presentation: problem statement & relevance; description of test 1 (trackers); description of SVM design. *Mid-semester report: administrative section (Gantt chart, list of roles & responsibilities); verification & validation (test 1); template setup. *HW4 (493): Business Lean - Problem, Customer Segments, & Early Adopters; Poster Draft - Background & Problem Statement. *Final report: template generation & formatting, problem statement, background/relevance, verification & validation: test descriptions (1 + 3), appendix: member roles/responsibilities. *Final presentation: problem Statement, testing/results/analysis (test 1+3)
Nico	*Programmed all model designs and tests. *Created & maintained GitHub. *Assists with file organization. *Consulted with Dr. Wei about ML and project approach. *Taught all group members how to use LabelMe. *Conducted initial research on ML, potential models, & programming tools. Selected project materials. *Wrote the previous version of the design history file organization document & initially set up the master file. *Wrote need & problem statement for background & market research assignment, revised need & problem statement for design requirements & standards assignment. *Ensures timely submission of deliverables in the absence of project manager. *Created prototype/demonstration video of model. *PDR presentation & report: tools/methods to generate design concepts; design alternatives (CNN + CNN); preliminary design. *CDR presentation & report: detailed design; preliminary design (CNN + CNN); future directions. *HW1 (493): test 3.

	*Labeled videos 1-23 with rectangle tracker. Labeled videos with polygon tracker: 1, 3, 8, 10, 14, 15, 27, 35, 49 *HW2 (493): ran tests; uploaded results (tests 1 & 2). *Mid-semester presentation: description of 3D CNN design; description of general testing methodology. *Mid semester report: verification & validation (general testing methodology); description of current design. *HW4 (493): Business Lean - Unique Value Prop, High Level Concepts, & Channels; Poster Draft - Design & Conclusion. *Final report: final design: algorithms/fabrication, verification & validation: methodology (programming environment, general procedure for testing & running the model). *Final presentation: final design (description & prototype)
Sedef	*Wrote background research section of background & market research assignment, and generated metrics for requirements & specifications assignment. *Assisted with the generation of potential model designs. *Assists with file organization. *Consulted with Dr. Wei about ML and project approach. *Picked up project materials. *Worked on model UI. *PDR presentation & report: alternative design (multilabel classification + CNN; random forest + ANN); requirements, metrics, & specifications chart. *CDR presentation & report: background & relevance; prototyping progress; design concepts (multilabel classification + CNN; random forest + ANN). *HW1 (493): test 3. *Labeled videos with polygon trackers: 7, 11, 12, 17, 18,21, 26, 29, 34, 36, 48, 51, 54 *HW2 (493): test 2. *Mid-semester presentation: description of tests (test 2); results & analysis. *Mid semester report: verification & validation (test 2). *HW4 (493): Business Lean - Existing Alternatives, Solution, Unfair Advantage, & Key Metrics; Poster Draft - Testing & Results, Discussion. *Final report: verification & validation: test 2, discussion. *Final presentation: testing/results/analysis: test 2, key results, limitations of final design.
Quynh	*Primarily facilitated external team communications (i.e., ASTEC, Dr. Dan, etc.) *Wrote market research section for background & market research assignment; wrote standards section for design requirements & standards assignment.

	*Ordered project materials from the BENG department. *PDR presentation & report: standards, requirements, metrics, & specifications; next steps. *CDR presentation & report: standards, requirements, metrics, & specifications; budget performance *HW1 (493): test 2. *Labeled videos with polygon trackers: 13, 16, 19, 23, 24, 28, 30, 31, 38, 46, 47, 50, 53 *HW2 (493): test 2. *Mid semester presentation: discussion; description of 3D CNN; budget update *Mid-semester report: upcoming test, next steps; administrative section (budget update). *HW4 (493): Business Lean - Cost Structure & Revenue Streams, Poster Draft - Requirements + Specs, Acknowledgements *Final report: requirements/specs/metrics/standards, executive summary appendix: budget, summary/conclusion. *Final presentation: project overview, requirements/specs/metrics/standards, final budget, conclusion/summary.
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Table B2. Extensive list of member responsibilities and tasks since Fall 2025.